MinSizeML: An R Package to Estimate the Minimum Sample Size in Supervised Learning for Classification.

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1. Introduction

Motivation One of the issues that should deserve greater attention in biomedical/clinical sciences research is the sampling process. To clearly define the target population, select the sampling frame, choose the sampling method, collect the data and assess the response rate are some of the steps required to conduct this process. However, the sampling process and downstream data analyses costs time, effort, money and of course carbon emissions [1], and all of these issues are based on the sample size estimate.

The sample size is the number of observations (e.g. participants, cells, units, etc.) that will be required to include in the study to answer the research hypothesis. If this value is too small, then it may not be able to answer the study question. But, if the number is too large, it may require substantial resources in terms of money, time and effort. Therefore, to find an optimum sample size is an essential aspect of any research [2].

Hypothesis While there are well known principles or considerations for sample size calculation in biomedical studies considering classical basic statistical approaches ([3] and [2]), the complexity and difficulty of calculating an appropriate sample size requires broad statistical knowledge, and well trained professionals in programming skills ([4] and [5]). Moreover, this task is not straight-forward when using machine learning approaches [6].

Objective In this work we have implemented an *R* package called *MinSizeML* based on the Figueroa *et al.* approach [7], where we have considered four supervised learning methods for classification: kNN, logistic regression, naive Bayes, and random forest. To assess the goodness of the classification, we fit a learning curve where the independent variable is the training set size, and the dependent variable is a metric, accuracy or Cohen's κ, obtained after each training step. The user defines a desired value for the metric and thus the corresponding minimum training size is returned from the learning curve.

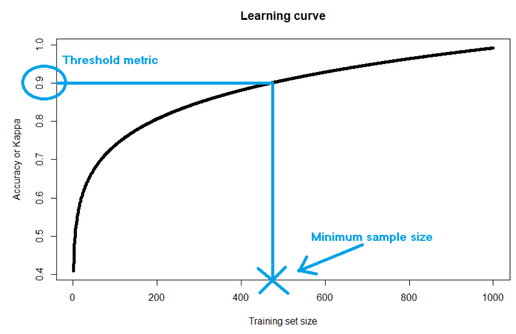
Availability *MinSizeML* is available on GitHub **https://github.com/gpcastelo/MinSizeML**,

2. Methods

1. Inputs: predictor variables X, variable Y to be predicted, and metric threshold.
2. Iteration through different values of the training step and implement the learning algorithm with *caret*[8]. Automatic division of the training and test size for each iteration. Metric calculation.
3. Fit the different training sizes and metrics to the **learning curve**, with the form of an inverse power law:

$$Y_{metric}(X_{size}) = (1-a) - b \cdot X_{size}^{c}, \qquad (1)$$

where,
   - $Y_{metric} \in [0,1]$, accuracy or Cohen's κ. These are defined as:
     - Accuracy $= \frac{correct\ \ predictions}{total\ \ predictions}$
     - Cohen's $\kappa = \frac{p_0 - p_e}{1 - p_e}$, where
       - p_0: observed agreement
       - p_e: chance agreement
   - X_{size}: training sample size.
   - $\{a, b, c\}$: parameters, obtained from fit with the *nls* package.
4. Threshold metric is used to calculate the **minimum training sample size**.

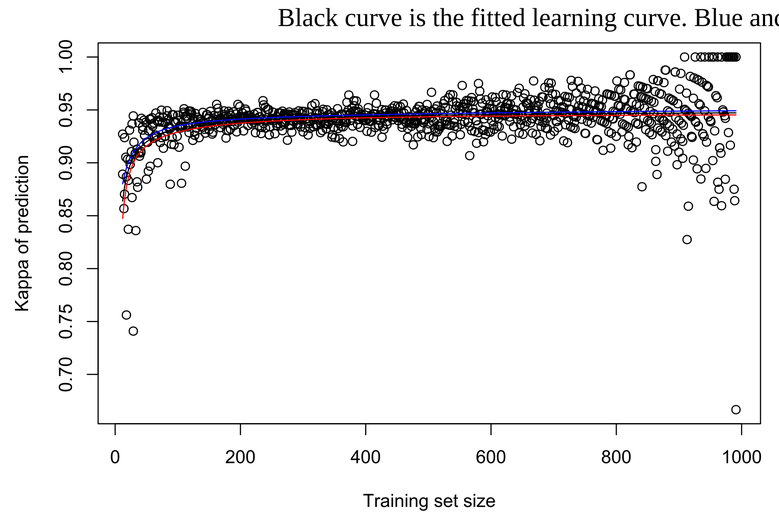


5. Moreover, our package has parallelism implemented out of the box (with the packages *doParallel* and *foreach*) so execution time may be sped up considerably.

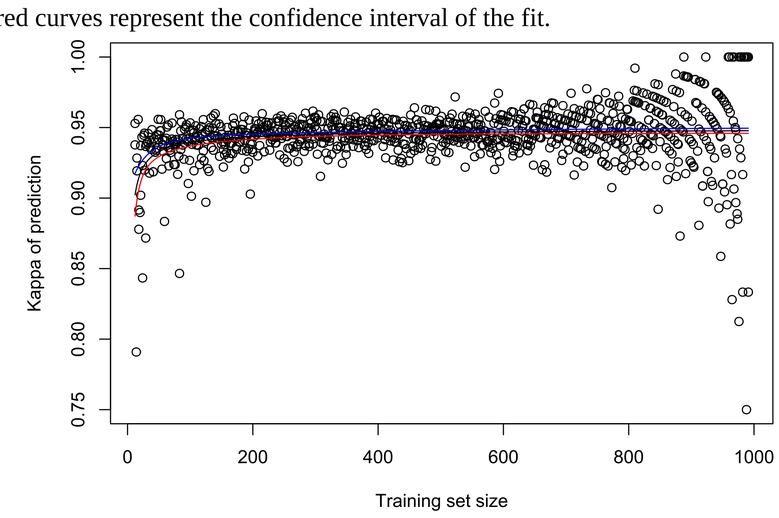
3. Results

Learning curves obtained from our package. Each circle represents a training size and its calculated metric.

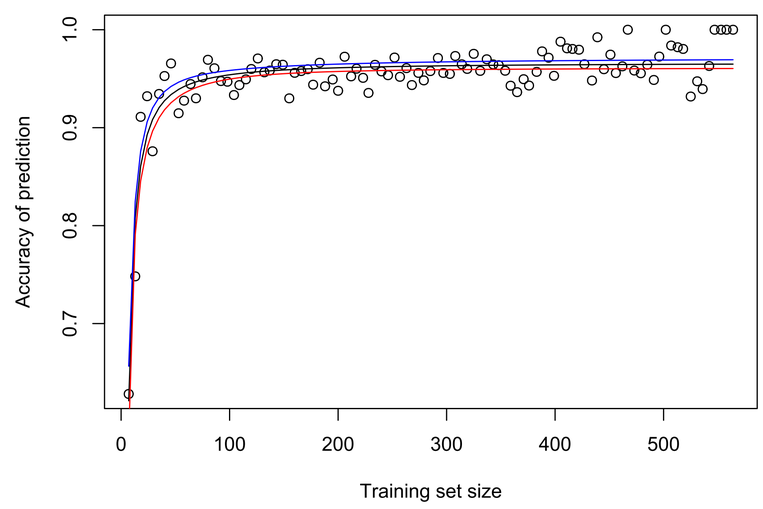
Black curve is the fitted learning curve. Blue and red curves represent the confidence interval of the fit.



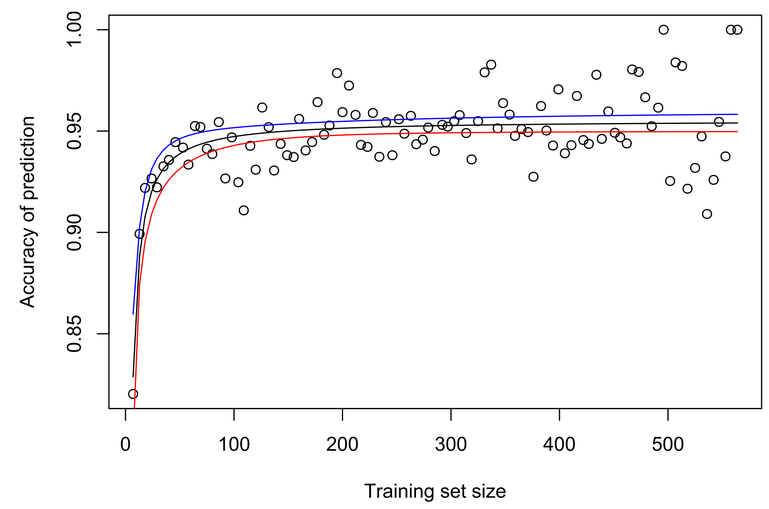
kNN evaluated with Cohen. Nmin = 25.3, for k = 0.9



Random forest evaluated with Cohen. Nmin = 11.2, for k = 0.9



kNN evaluated with accuracy. Nmin = 26.1, for acc = 0.9



Rando forest evaluated with accuracy. Nmin = 15.6, for acc = 0.9

Different data dimensionalities (x-axis) confirm the tendency of the learning curve.

The larger spread and downward tendency of the metrics for larger training sizes has a two-fold cause:

1. The larger the training size we use, the smaller the testing set size will be. As the testing size is decreased, the metric becomes less significant.
2. Overfitting: The downward tendency of the metric is clear when the training size reaches a certain value.

4. Conclusions

1. It is possible to predict the minimum sample size following the approach of [7], estimating metrics using equation 1.
2. There is a "sweet spot": once we reach a certain training set size, the calculated metrics do not improve much or even decrease with larger sizes.
3. We have created an *R* package that is open source and may be easily installed. This package allows for the systematic calculation of minimum sample size in different machine learning algorithms.
4. The formula for fitting metrics and training sample size is not always reflective of the measured metrics. Thus, the need for a more general method to estimate minimum sample size in machine learning problems for classification.

5. References

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6. Aknowledgments

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